



Fakultät Informatik

From CAD Software to Slicer

Workflow Optimization using Print Hints in 3MF Files for 3D Printing

Bachelor Thesis Exposé
im Studiengang Medieninformatik

vorgelegt von

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1 Introduction

It is one of the clearest paradoxes of modern work that computer-based systems designed to reduce task complexity and cognitive workload actually often impose even greater demands and stresses on the very individuals they are supposed to be helping.[4, p. 222]

When designing a model for 3D printing, the process frequently follows a certain set of iterative steps. After the ideation phase, the design engineer first creates the model using CAD software, imports the file (e. g. STL, 3MF) to a slicer software, sets printing parameters and finally starts the printing. The result is then evaluated for design flaws or areas of improvement, leading to model revisions and thereby reinitiating the workflow. Despite automation and optimization having been a prevalent topic in the software developer community for years, some of these repetitive steps have the potential to become more efficient and user friendly.

This thesis aims to prove the feasibility of ‘print hints’ as a way to improve the workflow and thus reducing cognitive workload. The approach as currently planned contains two core components: firstly, an improved export function within the CAD software *Autodesk Fusion*, e. g. to mark certain objects as modifiers; and second, automatic handling of materials and these print hints within the open source slicer software called *OrcaSlicer*.

2 Description

2.1 Definition of Problem

As mentioned above, the process of designing and printing 3D models either for private purposes or as prototyping in a professional context frequently involves repetitive steps, in particular when importing the 3D model to the slicer software. Each time the model is imported, the user has to manually set the printing parameters. In some cases, this can be as easy as selecting the filament and continuing with the standard presets but more often, one or several of the following steps are involved as well:

- assigning different filaments to different model parts
- setting support blockers or enhancers
- setting modifiers, e. g. for perimeters

Such repetitive tasks not only increase the cognitive workload but also reduce process efficiency, potentially leading to higher error rates and waste of resources.

2.2 Research topic: proof of concept

The bachelor thesis aims to prove the feasibility of exporting print hints within the 3MF file from *Autodesk Fusion* and have the slicer automatically parse them to automate the workflow, reducing the need for repetitive manual steps.

2.3 Solution approach

Based on the current knowledge, the solution will consist of multiple parts.

1. It should be possible to add print hints via Autodesk Fusion's user interface. Since the software is not open source, the only way to add this functionality would be as a script or plugin. Autodesk Fusion provides a comprehensive amount of methods within their API for developers to use.

2. The slicer should then be able to detect the different materials from exported 3MF files and automatically map them to filament presets, as well as automatically handle print hints. As mentioned before, *OrcaSlicer* is open source, therefore the changes can be directly integrated into a forked repository.
3. Furthermore it needs to be determined whether print hints can be written to the 3MF file's XML data containing the 3D models without changing the file's standard.

2.4 Research methods

Since the goal is a proof of concept, the methods used will mainly be to create software prototypes. Additionally, the written part will illustrate the design decisions as well as any other important steps, serve as documentation and give the reader foundational knowledge regarding the design process for 3D models and the working of slicer software.

Supporting both parts, literature research will be necessary. The most important literature regarding the implementation will be Autodesk's Fusion API documentation[1], the documentation of the OrcaSlicer[2] as far as available and up to date, and 3MF Consortium's file specifications[3]. Works on human-computer interaction, such as [4], as well as studies on 3D printing and the maker community, like [5] and [6], will provide a deeper understanding and context for the problem at hand.

2.5 Current state of research

Insights so far:

- Fusion exports materials as hexcode colours to the 3MF file where they are being referenced in each 3D object but not imported by the OrcaSlicer yet
- 3MF saves 3D models in easily human-readable XML format which is therefore easily edited
- OrcaSlicer doesn't throw any errors when importing manually edited 3MF files
- Fusion provides a comprehensive API enabling the development of plugins or scripts for extended functionality

3 Planned timeline

Until	Milestone
mid-June	• literature research • further code analysis of the OrcaSlicer • implementation: automatically mapping material presets to material colours in 3MF file import • first draft of the introduction and problem analysis
mid-July	• create plugin to add print hints to the 3MF file when exporting models from Fusion • first draft of theoretical foundation regarding slicer and the solution approach
mid-August	• handling print hints within OrcaSlicer • first complete draft
end of August	• revisions, finalizations and quality control

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